Calorimeter pointing: v4 Stage 1 preparation

Stored inputs, offline smearing and the bounded signal check

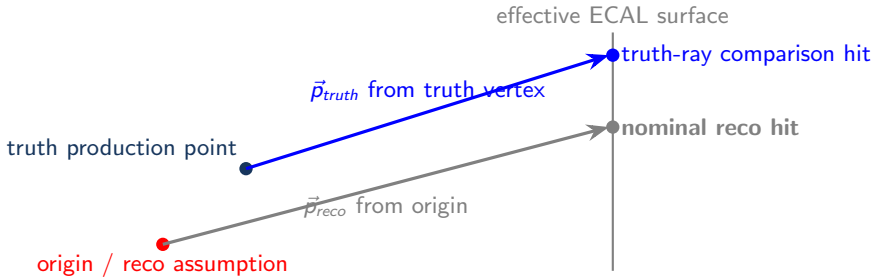
FCC-ee $\Lambda_b \to \Lambda\gamma$ study

5 October 2026

PI checkpoint: prepare reproducible Zbb, Zcc, Zss and 100k forced-mode reprocessing.

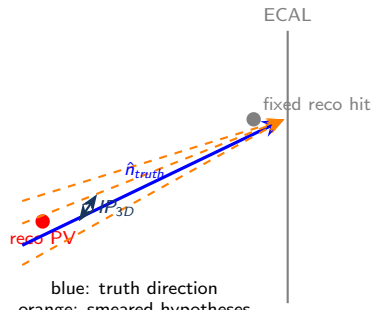
Procedure, steps 1–3: build the nominal hit

1. **Read Stage 1 candidate fields.** Take the reconstructed photon energy and momentum, its matched truth photon momentum and production point, the reconstructed PV, and the effective ECAL surface parameters. The photon energy response is already applied in Stage 1.
2. **Choose the nominal hit anchor.** Project the reconstructed photon momentum, whose direction assumes production at the origin, from $(0, 0, 0)$ to the first forward crossing of the effective cylinder or endcap. This saved `reco_hit` is the hit position used by the nominal offline study.
3. **Keep a truth-ray comparison.** Independently project the matched truth momentum from the truth production point. This `truth_hit` is useful for geometry and closure checks; it does not replace the nominal reco-derived hit.



Procedure, steps 4–6: emulate pointing and measure displacement

4. **Attach the truth direction to the reco hit.** Normalize the matched truth momentum to obtain $\hat{n}_{truth}$. The truth production point sets neither the nominal hit nor a new photon origin.
5. **Make a resolution hypothesis.** Draw two Gaussian offsets in orthogonal directions perpendicular to $\hat{n}_{truth}$; combine them with that direction and renormalize. Repeat for per-component $\sigma = 0.5, 0.7, 0.9, 1.1, 1.3, 1.5$ mrad. Reuse deterministic draws across hypotheses and candidates sharing a photon.
6. **Calculate the displacement.** Set $\vec{p}_{hyp} = E_{reco}\hat{n}_{smear}$. Use the fixed reco hit and reconstructed PV to compute $IP_{3D} = |(\vec{x}_{reco\ hit} - \vec{x}_{PV}) \times \hat{n}_{smear}|$ and transverse $|d_0|$. Apply and scan cuts offline after BDT and the chosen other selections.



blue: truth direction
orange: smeared hypotheses

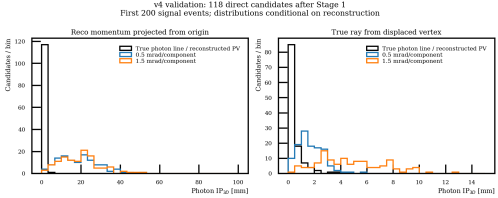
Across hypotheses only the direction changes. Energy, reco hit, PV and candidate list stay fixed; no hit-position smearing or covariance is included.

Saved inputs and pilot geometry check

For each reconstructed Λ_b candidate the v4 tuple keeps:

- measured photon $E_{\mathrm{reco}}, (p_x, p_y, p_z)_{\mathrm{reco}}$ and reco index;
- matched truth photon (p_x, p_y, p_z, E) and production vertex;
- predicted ECAL hit from the displaced truth ray and, separately, from the current origin-based reco direction;
- reconstructed PV and effective surface parameters.

Smear the truth direction in two orthogonal tangent directions with Gaussian component width $\sigma = 0.5$–1.5 mrad (six points), then set $\vec{p}_\gamma^{\,hyp} = E_{\mathrm{reco}}\hat{n}_{\mathrm{smear}}$. Reuse the same random draw for a photon shared by candidates. Evaluate $IP_{3D} = |(\vec{x}_{hit} - \vec{x}_{PV}) \times \hat{n}|$ and $|d_0|$ offline after BDT.



118 matched direct candidates from the first 200 signal events. Left: hit from the origin projection of reconstructed momentum. Right: hit from the displaced truth ray. Both smear the truth direction; shapes are conditional, not yields.

Effective IDEA geometry: barrel $R = 2250$ mm, $|z| = 2500$ mm; endcap inner radius inferred as $2500/\sinh(3)$. A pointing significance needs a position/direction covariance not yet modelled.

What the 200-event signal check established

Input events	200	Candidate-bearing events	121
Candidates	133	Direct / wrong combinations	118 / 15
Old branches identical	474 / 474	Pointing vectors valid	28 / 28
Stage 2 audit / selected	133 / 129	After frozen BDT cut	46

What passed

The true-ray projection closes to its saved hit. The new fields survive Stage 2 preparation and the frozen BDT score. The paired run kept the same 121 output events and the same 133 candidates.

What remains to measure

Full signal and Zss/Zcc processing must establish unresolved fractions, pointing-cut efficiencies, mass and angle sculpting, and signal acceptance. This small signal check does not measure flavour rejection.

v4 Condor check only

Zbb: 4,398 files; Zcc: 5,018; Zss: 5,015. Each has 1,200 chunks, four cores/job, accounting group `group_u_FCC.local_gen` and `workday` flavour. All three checks passed; no jobs were submitted.

100k Stage 1 rerun: available modes and next commands

Mode	Current merged input	Status
$\Lambda\gamma$ PHSP	`Lb2LambdaGamma_nev100000...root`	ready
$\Lambda\gamma$ HELAMP	`Lb2LambdaGammaPhysics_nev100000...root`	ready
$\Lambda\eta$ PHSP / HELAMP	`Lb2LambdaEta(Physics)_nev100000...root`	ready
$\Lambda\pi^0$ HELAMP	`Lb2LambdaPi0Physics_nev100000...root`	ready
$\Lambda\pi^0$ PHSP	`Lb2LambdaPi0_nev100000...root`	merged file not found

1. Run `scripts/run_stage1_v4_forced_100k.sh --check-only`; it lists exact inputs/outputs. Then use `--run` to process available modes. Default uses four concurrent samples $\times$ eight CPUs (32 cores); tune `V4_DIRECT_JOBS` and `V4_DIRECT_CPUS`. Each output gets a v4 tuple validation JSON.
2. Generate/merge the missing π^0 PHSP 100k sample using seed base 72001 from `config/config.yaml`, then rerun the checker.
3. After review, submit Z flavours with `scripts/submit_stage1_v4_zflavours.sh --submit`; default is `--check-only`.

Production commands, output paths, file inventory and check logs are recorded in the v4 howto and frozen job preparation note. No full 100k or Condor production was launched here.